

SIKSHA 'O' ANUSANDHAN (DEEMED TO BE UNIVERSITY)

Faculty of Engineering & Technology | Department of Computer Science & Engineering

PRACTICAL ROBOTICS PROJECT WITH ARDUINO (CSE 4571)

LAB REPORT – INDEX

Name:		Regd. No.:		
Section:		Semester:	CSE 7 th	
Sl. No.	Aim of the Experiment	Date	Page No.	Faculty Signature
1	To interface an HC-SR04 ultrasonic sensor with an Arduino UNO, measure object distance, and implement threshold-based obstacle detection using LEDs and a buzzer.			
2	To interface an analog sensor with Arduino UNO, acquire ADC data and study its response to changing illumination.			
3	To control a micro-servo using a potentiometer as a position command.			
4	To interface a bipolar stepper motor with an A4988 driver and control its position and direction.			
5	To integrate a PIR motion sensor, servo and buzzer to create an automatic robotic response.			
A1	Assignment -1			
6	To implement a responsive robotic controller using an HC-SR04 distance sensor, a DC motor, a pushbutton and a buzzer, with a finite state machine and millis()-based timing.			
7	To implement a simple feedback-based servo position control system using an Arduino UNO, a setpoint potentiometer, an external feedback potentiometer and an SG90 micro servo.			
8	To develop an understanding of Object-Oriented Arduino Programming and design reusable robotic classes for simple robotic components, and to integrate these concepts to develop a modular robotic application.			
9	To interface an IR receiver with an Arduino UNO, decode NEC protocol signals from an IR remote, and control a servo motor (robotic actuator) based on the received commands.			
A2	Assignment - 2			
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